



3009ICT

Data Processing and Visualisation

Assignment (70%)

Topic: Data Analysis on Car Advertisements.

The objective of this assignment: Utilise a dataset of car sales to derive insights into factors influencing car prices and profile the sellers.

Submission: Submit your Jupyter Notebook file, which must contain all of your work and analysis. Include your explanations and discussion in text cells within the notebook. Before submission, re-run every cell to ensure there are no errors and that all figures and results are clearly displayed. For the interactive visualisation component, you may use other tools, but you must include the link to your interactive visualisation in a text cell at the end of the submitted notebook. You may also include other supporting evidence, such as a deployed web page link or a video demonstration, if relevant. Students may use Generative AI tools to assist learning and debugging. However, all submitted work must represent the student's own understanding. During the interview, students must explain and justify all submitted code and analyses independently. All work must be your own.

Total Points: 35. **You need to achieve >40% of this assessment to pass the course.**

Note: Near the end of the course, an individual interview may be conducted to assess each student's understanding of the assignment. The interview will also be worth 35 points and will count toward the final assignment mark.

Download Data:

In this assignment, we will analyse the dataset on car sales advertisements. There are two parts of the data, which are datasets on cars currently on market and the information of the sellers. Use these links to download the datasets:

- **car_dataset.csv:** https://drive.google.com/file/d/1H5_SQGLPMIGnSSs3mFIAipt-Q5KYJkb_/view?usp=sharing
- **seller_dataset.csv:**
<https://drive.google.com/file/d/1YrqGTSzek0Bdful8ysraGfN8fFPurSzB/view?usp=sharing>

ad_id	origin	condition	car_model	mileage	exterior_color	interior_color	num_of_door	seating_capacity	engine
17042	Domestic as	New car	Truck	0 Km	White	gray	2-door	2-seat	Petrol1.0 L
53794	Imported	New car	SUV	0 Km	Black	Black	5-door	7-seat	Petrol3.4 L
73954	Domestic as	New car	Crossover	0 Km	Silver	Brown	5-door	8-seat	Petrol2.0 L
74150	Imported	New car	SUV	0 Km	White	Black	5-door	5-seat	Petrol1.8 L
87573	Domestic as	New car	Crossover	0 Km	Silver	gray	5-door	8-seat	Petrol2.0 L
97011	Domestic as	New car	Van/Minivan	0 Km	White	gray	5-door	2-seat	Petrol1.0 L
101726	Domestic as	New car	SUV	0 Km	White	Black	5-door	7-seat	Petrol1.5 L
135739	Imported	New car	SUV	0 Km	Copper	Black	5-door	7-seat	Petrol2.7 L
142495	Domestic as	New car	4x4	0 Km	Grey	Black	4-door	5-seat	Diesel2.2 L
143308	Domestic as	New car	4x4	0 Km	Black	Black	4-door	5-seat	Diesel2.2 L
174951	Domestic as	New car	Van/Minivan	0 Km	White	Cream	4-door	7-seat	Diesel2.2 L
182003	Domestic as	New car	Crossover	0 Km	Silver	gray	5-door	8-seat	Petrol2.0 L
182298	Imported	New car	SUV	0 Km	White	Cream	5-door	5-seat	Petrol2.7 L
183963	Domestic as	New car	Sedan	0 Km	White	Yellow	4-door	5-seat	Petrol1.5 L

ad_id	name
17042	Suzuki - Sydney
53794	Toyota - Melbourne
73954	Toyota - Brisbane
74150	Toyota - Perth
87573	Toyota - Adelaide
97011	Suzuki - Canberra
101726	Honda - Darwin
135739	Toyota - Hobart

1. Data Formatting:

Note: The cleaned dataset is provided as an example. You are expected to write your own code to perform the following formatting tasks. Direct use of the provided cleaned dataset without applying the required coding steps will result in **mark deductions**.

All tasks below are for dataset **car_dataset.csv**.

- a. Numerical Conversion: Convert 'mileage', 'num_of_doors', 'seating_capacity', 'fuel_consumption', and 'price' to numerical values by removing any text and ensuring the correct data type.
- b. Engine Feature Split: Split the 'engine' column into two new columns: 'type_of_engine' (e.g., Petrol, Diesel) and 'engine_capacity' (numeric value representing engine size). At the end of this task, you will have two new columns: 'type_of_engine' and 'engine_capacity'. Remember to remove Litres (L) in engine_capacity.
- c. Categorical Encoding: Apply one-hot encoding or label encoding to 'origin', 'condition', 'car_model', 'exterior_color', 'interior_color', 'type_of_engine', 'fuel_system', 'transmission', 'drive_type', 'brand', and 'grade'. The result of this can be used in the following section for data visualisation.
- d. Standardising text data: Converting all entries to lowercase in these columns: 'origin', 'condition', 'car_model', 'exterior_color', 'interior_color', 'type_of_engine', 'fuel_system', 'transmission', 'drive_type', 'brand', and 'grade'.
- e. Numerical Scaling: Normalise all processed numerical variables, including the newly created 'engine_capacity', 'num_of_doors', 'seating_capacity', 'fuel_consumption', and 'price' using min-max normalisation. Remember to save it in new columns, not original columns.

E.g., 'engine_capacity' → new column named: 'normalised_engine_capacity'

Note: If a student uses the cleaned example dataset without demonstrating the coding process, there will be a deduction of half the points assigned to each task where the coding step is skipped. For example, if a student does not code the numerical conversion and uses the cleaned data instead, they will receive 1 out of 2 points for that task.

2. Data Exploration and Cleaning:

- a. Statistics Computation: For the **car_dataset.csv**, Calculate basic statistics (mean, median, mode, standard deviation, range) for the variables 'price', 'mileage', and 'year_of_m'.
- b. Outliers and Missing Values:
 - i. Detect outliers using the IQR method and justify whether each outlier should be removed, retained, or transformed.
 - ii. Identify and treat missing values across the dataset. Choose appropriate methods for imputation (mean, median, mode, or predictive modeling) based on the variable type. Students must justify why the selected imputation method is appropriate for each variable. Remember to show how you deal with the outliers and missing values. The results before and after should also be included. E.g.,

Before cleaning data: 150000 rows. After cleaning data: 113452 rows.

3. Data Storage:

Save the processed data above to a new CSV file, naming it: “**processed_car_detail_en.csv**”. Using library **sqlite3** in Python, conduct these tasks:

- a. *Cars* Table: Create and store a ‘Cars’ table containing car advertisement details after cleaning and formatting. Ensure they have correct data types, and include only those specific fields in the table:

- ad_id - INTEGER (set this as Primary key)
- car_name - VARCHAR(255)
- origin - VARCHAR(255)
- condition - VARCHAR(255)
- exterior_color - VARCHAR(255)
- car_model - VARCHAR(255)
- year_of_manufacture - VARCHAR(255)
- transmission - VARCHAR(255)
- price - INTEGER
- drive_type - VARCHAR(255)

- b. *Sellers* Table: Create table ‘Sellers’ containing information of the seller to the ads using ‘**seller_dataset.csv**’. Ensure they have correct data types, and include only those specific fields in the table:

- ad_id - INTEGER (set this as Primary Key)
- name - TEXT
- address - TEXT
- website - TEXT
- phone1 - INTEGER

- c. Make a query that returns the output below:

ad_id	Cars.car_name	Cars.price	Sellers.name	Sellers.phone1
17042	Suzuki Super Carry Truck 1.0 MT 2022	249,000,000	Suzuki Viet Anh	972251486
...	...	...	...	...

If a SQLite database is successfully implemented with the above requirements, full marks are awarded.

Note: If the student is unable to store the data in a SQLite database, the data must then be saved into .csv files with the same 'cars' and 'sellers' table structure. However, choosing this option means no points will be awarded for this task, as the learning objective is to practice database storage and management in the database.

4. Data Visualisation:

a. Static Visualisation: For each of the following requirements, you need to select an appropriate chart. Marks will be deducted for using inappropriate chart types (deduction applied for each inappropriate chart) or duplicated chart types (deduction applied for each duplicated chart).

- i. Examine the relationship between 'mileage' and 'price' . Analyse the trend of car prices over the 'year_of_m'.
- ii. Compare the average prices among different 'car_models'.
- iii. Display the distribution of 'engine_capacity'.
- iv. Visualise the distribution of the cars' exterior colors as the percentage of a whole.
- v. Visualise/Identify correlations between these features: 'mileage', 'num_of_doors', 'seating_capacity', 'fuel_consumption', and 'price'.

Each chart should be clearly labeled with titles, axis labels, and a legend. *Hint: You can use **Seaborn** or **Matplotlib** to generate the figures.*

b. Interactive Visualisation Enhancement:

Enhance two of the visualisations created in task 4 with interactive elements. This could include but is not limited to, hover details, clickable legends, sliders, or dynamic filters. If the interactive elements are not implemented, you will not receive any marks for this task.

End of assignment questions

Marking Criteria

A. Submission (Notebook): 35 points		
Data Formatting (10 points)	Data type conversion and cleaning accuracy	2 points
	Feature engineering from raw fields	2 points
	Standardisation and consistency of transformed data	3 points

	Scaling logic and reproducibility of preprocessing workflow	3 points
Data Exploration and Cleaning (10 points)	Statistical computation, interpretation, and explanation	5 points
	Check and address outliers with justified method	2 points
	Identify and handle missing values with justification	3 points
Data Storage (5 points)	Create Cars table with correct schema	2 points
	Create Sellers table with correct schema	2 points
	Make the specified query and show correct output	1 point
Data Visualisation (10 points)	Relationship analysis between 'mileage' and 'price': appropriate visual choice, explanation, and readability	1 point
	Trend of car prices over the 'year_of_m': appropriate visual choice, explanation, and readability	1 point
	Compare average prices among different 'car_models': appropriate visual choice, explanation, and readability	1 point
	Distribution of 'engine_capacity': appropriate visual choice, explanation, and readability	1 point
	Distribution of exterior colours as percentage of the whole: appropriate visual choice, explanation, and readability	1 point
	Correlations between specified features: appropriate visual choice, explanation, and readability	1 point
	Interactive Visualisation 1: meaningful interactivity, usability, and analytical value	2 points
	Interactive Visualisation 2: meaningful interactivity, usability, and analytical value	2 points
B. Individual Interview (Online in Week 12): 35 points		
Understanding of Data Formatting		7 points
Understanding of Data Cleaning		7 points

Understanding of Data Storage	5 points
Understanding of Data Visualisation	7 points
Understanding of Code and Analysis	9 points